Digital Clock

Chaitanya and Vinay
ee26btech11201 and ee26btech11233

Progress log

Introduction

This document is made for recording the progress done with our project of creating a digital clock .

This project is to create a functional clock with additional features such as alarm, editing the time, and also make sure that there are no errors in the functioning of the clock.

Materials

Arduino Uno , IC 7447, breadboard , 4 switches, 7 of 220 ohm resistors, 4 of 12 kilo ohm resistors, 6 of Common Anode 7-segment displays, 80 of Jumper wires.

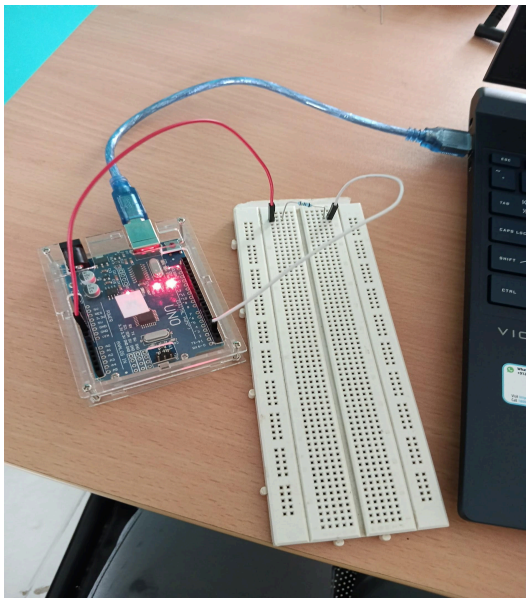
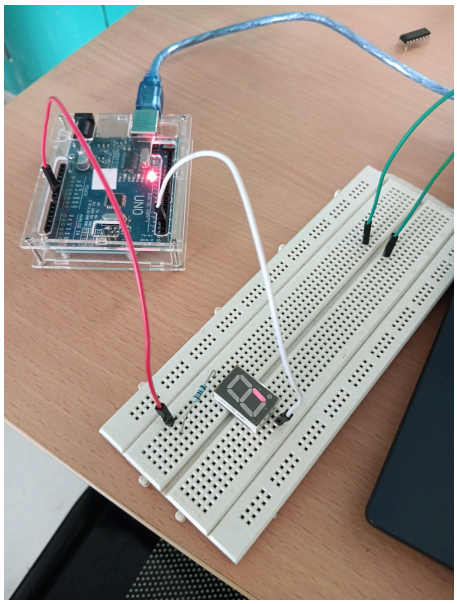
Learning

We first need to know what the actual aim is for the project. After that we need to learn what each of the components in the circuit do, how to connect each component safely so that they won't get damaged permanently. Also we need to learn the code which is provided.

Progress log
– We had an interactive session with our seniors , who explained the concepts of K-Maps for us and also explained the project details.

– We first tested if everything we received is working correctly as intended and there are no faulty components. We tested it with a simple arduino code that we got from AI. It basically is a code that shows whether the circuit is closed or open. It will show the output in the serial

monitor , switching between on and off every second. We tested individual components one by one and found that all the parts are in proper condition. We also tested all the rows in the breadboard to make sure that we don't run into troubles later during the project.

https://youtube.com/shorts/Pgmq20Qlu6M?feature=share

– We tested the IC 7447 by using an AI generated code ,which simply showed all the digits on a 7-segment display one by one. We connected all the jumper wires as told by AI and we initially had a lot of trouble getting it to work as intended. But finally we got it to work by working a lot about the possible failure points.
https://youtube.com/shorts/_SZHmFRoUtk?feature=share
– We made all the K- Maps corresponding to all the digits and we double checked our results to make sure that this would not become a problem.

The tables provided in the next page are the logic tables for incrementing different digits by 1

Truth table for seconds and minutes:

INPUT					OUTPUT			
Z	Y	X	W		D	C	B	A
0	0	0	0		0	0	0	1
0	0	0	1		0	0	1	0
0	0	1	0		0	0	1	1
0	0	1	1		0	1	0	0
0	1	0	0		0	1	0	1
0	1	0	1		0	1	1	0
0	1	1	0		0	1	1	1
0	1	1	1		1	0	0	0
1	0	0	0		1	0	0	1
1	0	0	1		0	0	0	0

XW ——-- ZY	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	-	-	-	-
10	1	0	-	-

(a) A = W1′

XW ——-- ZY	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	-	-	-	-
10	0	0	-	-

(b)=W1X1′Z1′+W1′X1

XW ——-- ZY	00	01	11	10
00	0	0	1	0
01	1	1	0	1
11	-	-	-	-
10	0	0	-	-

XW ——-- ZY	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	-	-	-	-
10	1	0	-	-

(c)C = X1′ Y1 +W1′ Y1 +W1X1Y1′ (d) D =W1′ Z1 +W1X1Y1

Figure : counting 0-9

Truth table for seconds-tens and minute-tens:

Z	Y	X	W		D	C	B	A
0	0	0	0		0	0	0	1
0	0	0	1		0	0	1	0
0	0	1	0		0	0	1	1
0	0	1	1		0	1	0	0
0	1	0	0		0	1	0	1
0	1	0	1		0	0	0	0

XW ——-- ZY	00	01	11	10
00	1	0	-	1
01	1	0	-	-
11	-	-	-	-
10	-	-	-	-

(a) A = W2′

XW ——--- ZY	00	01	11	10
00	0	1	0	1
01	0	0	-	-
11	-	-	-	-
10	-	-	-	-

(b) B =W2X2′ Y2′ +W2′ X2

XW ——-- ZY	00	01	11	10
00	0	0	1	0
01	1	0	-	-
11	-	-	-	-
10	-	-	-	-

(c) C = W2X2+W2′ X2′ Y2

Figure : counting 0-5

Truth table for hours:

X	W		D	C	B	A
0	0		0	0	0	1
0	1		0	0	1	0
1	0		0	0	1	1
1	1		0	0	0	0

W ——— X	0	1
0	1	0
1	1	0

(a) A = W5′

W ——— X	0	1
0	0	1
1	1	0

(b) B =W5X5′ +W5′ x5

Figure : Counting 0-3

Truth table for hours-tens:

X	W		D	C	B	A
0	0		0	0	0	1
0	1		0	0	1	0
1	0		0	0	0	0

X —-- W	0	1
0	1	0
1	0	-

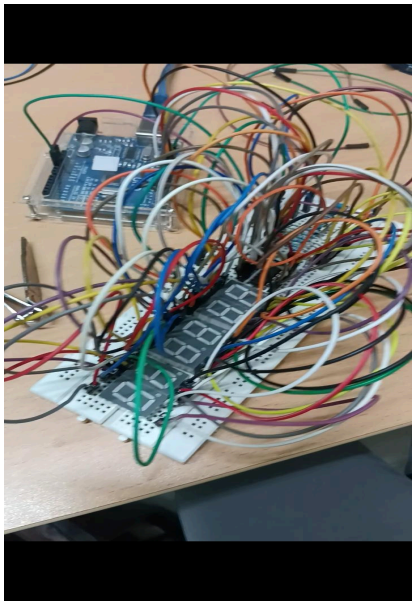
(a) A = W6′ X6′

X —-- W	0	1
0	0	1
1	0	-

(b) B =W6X6′

Figure : counting 0-2

– We now wanted to see everything in action at once. We wanted to build the clock as intended by the manual. This would help us get much information about the code and we can tweak the code to see how it impacts the output.



https://youtube.com/shorts/DHVo1KJDS3c?feature=share

We asked AI for connection details and it gave us some instructions. We first put the 7-segment displays and IC 7447in place and we started connecting them one by one with the jumper wires. We connected all the wires and when we connected the arduino and uploaded the original code , to our disappointment, it was working really bad. Almost all the digits are not functioning properly and there is some weak glow in some digits.

After some careful inspection of every wire, we found some wires that were wrongly connected. We reconnected it back to correct positions and ran the code again.
Still it was having some problem with the display. We searched about this problem for a really long time. Finally, we resolved the problem a little bit by connecting wires in a better way. We found that there are some faulty sockets in the breadboard.

–We began to read the code line to line and try to understand it with the help of AI. We learnt a lot about the code and the logic behind the increment of digit and also a lot about the different functions and when they activate.

– We found a surprising fact when we were reading about the code. We learnt that there is no need for the 12 kilo ohm resistors. We are using internal pull_up resistors. There is no need for another external resistor.

–We finished going through the original code and we wanted to learn more . So we asked AI for a code that has a similar function as the original code but does not use BCD to update time. This code is much simpler and we learnt this too. After going through both the codes, we now had much more idea about how things work.

–After thoroughly going through the code, we found a flaw with the original code. When we go into a pause state, and we change the hours into 8 or 9 and then when we change the hours-tens digit, we can actually get the time set at 28 or 29 hours, which is obviously not true. For this we came up with a simple fix such that it will not increment the hours-tens digit beyond 1 when the hours digit is between 4 and 9.

–When other people (e.g:family) sees this clock with 80 jumper wires, they say “Why is it so complex?”. We thought about it and Chaitanya remembered about his previous arduino experience and we had the idea of the solid core wires. This clock looks so tightly packed because of the jumper wires. We had to connect sockets that are closely spaced apart with an entire jumper wire. This made the circuit very ugly and full of wires which made seeing the display difficult and also made finding the fault very tedious.
So we went back to the core labs and asked for solid core wires. Fortunately they had some used wires which we brought back to the room. We carefully cut a few wires to desired sizes and started connecting the components with a better layout for the wires so that they are easy to connect and less bundled. We connected the wires coming from the arduino with jumper wires so that it has some flexibility to move.

–We connected everything and uploaded the original code to make sure everything works. We initially misplaced a wire and caused an error. Later we saw a problem. Segment f of all displays is not working. We checked every wire connecting those f segments and figured out that there is a faulty line in the breadboard connecting the IC.

–We decided to start to modify the original code such that it has an alarm feature.
We removed the decrement button from the code and changed it into the alarm button.
The alarm feature is such that when you set the alarm and the alarm time matches the clock time, the display is changed to display alternating 8 and 0.

– We continuously update the code and test the features in the arduino side by side. In this way we know where exactly the change is required and tweak it and test it again to see if the problem is fixed. Sometimes, there would be a problem that we could not correct for so long(showDigit function not being declared). In that case we took help of AI to solve the problem and understand more about the program language.

– We also made some minor changes to the code such as whenever you increment the digit, it first stays lit so that you can see what the number is when you are rapidly switching the numbers.

–Finally we finished the code in the way we wanted it to be and implemented the alarm feature along with the clock. The clock now looks really good compared to how it looked with only jumper wires. There is ample space for seeing through the connections and see if there are any faulty connections.
https://youtube.com/shorts/iV97-8nE6mg